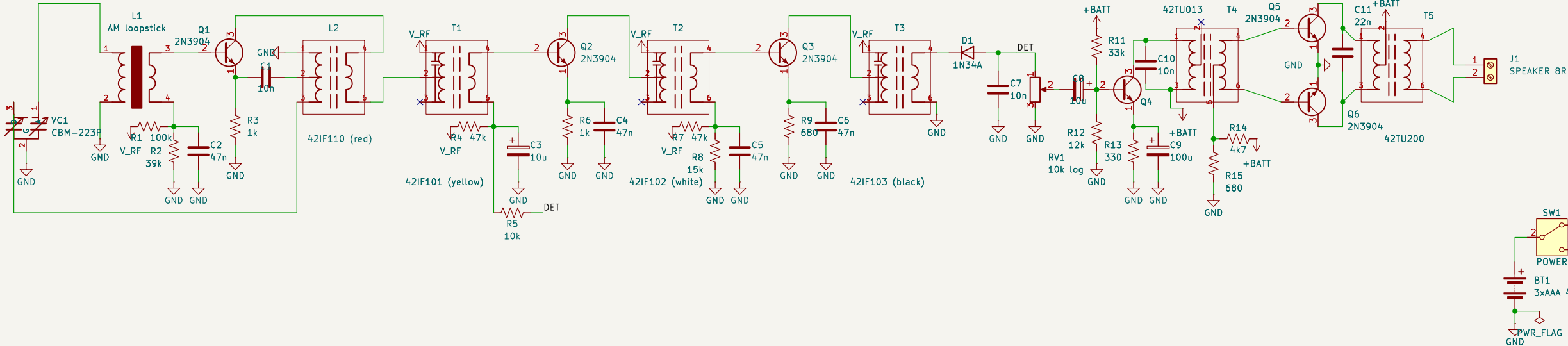


AM BROADCAST RECEIVER – 6 transistor superheterodyne, 455 kHz IF, 4.5 V (3xAAA), negative ground, all 2N3904 NPN

Stages: Q1 converter (mixer + local oscillator, L2 red) -> T1 (yellow) -> Q2 IF amp with AGC -> T2 (white) -> Q3 IF amp -> T3 (black) -> D1 detector -> RV1 volume -> Q4 driver -> T4 -> Q5/Q6 class-B push-pull -> T5 -> 8 ohm speaker



Alignment: set VC1 fully closed, adjust L2 slug so the local oscillator is ~985 kHz (station at 530 kHz); at the high end use the VC1 oscillator trimmer for ~2055 kHz (1600 kHz). Peak T1, T2, T3 slugs for max audio on a weak station (they are pre-tuned to 455 kHz). Peak L1 by sliding the coil on the rod at the low end and the VC1 ante

AGC: D1 delivers negative-going audio + DC. R5/C3 filter it onto the Q2 base bias node; strong stations reduce Q2 bias and gain. V_RF is the 330R/100u decoupled rail for the RF/IF stages. Electrolytics: C3, C9, C12, C13 + up; C8 + toward Q4 base.

Substitutions: 2N2222/BC547 for 2N3904 (check pinout). 1N60/BAT46 for 1N34A. T1..T3 and L2 are Xicon 42IF101/102/103/110 (10 mm cans). T4 = Xicon 42TU013 (1k CT : 8 CT). T5 = Xicon 42TU200 (200 CT : 8). VC1 = 2-gang polyvaricon 140/60 pF with trimmers.

455 kHz IF, 4.5 V, all NPN 2N3904, negative ground

Sheet: /
File: am-radio.kicad_sch

Title: AM Broadcast Receiver – 6 transistor superhet

Size: A3 Date: 2026-09-11

Rev: 1.0

KiCad E.D.A. 9.0.2

Id: 1/1